

Module Title: Routing Concepts

Module Objective: Explain how routers use information in packets to make forwarding decisions

Topic Title	Topic Objective
Path Determination	Explain how routers determine the best path.
Packet Forwarding	Explain how routers forward packets to the destination.
Basic Router Configuration Review	Configure basic settings on a router.
IP Routing Table	Describe the structure of a routing table.
Static and Dynamic Routing	Compare static and dynamic routing concepts.

Page #	Activity Type	Activity Name
14.1.7	Check Your Understanding	Path Determination
14.2.4	Check Your Understanding	Packet Forwarding

14.3.5	Packet Tracer	Basic Router Configuration Review
14.4.13	Check Your Understanding	IP Routing Table
14.5.6	Check Your Understanding	Dynamic and Static Routing

New Terms

best path

longest match

prefix length

next-hop router

process switching

fast switching

Cisco Express Forwarding (CEF)

route sources

static routes

dynamic routing protocols

ip route

Default Route

ip route 0.0.0.0 0.0.0.0 [exit-if | next-hop-ip]

ipv6 route ::/0 [exit-if | next-hop-ipv6]

Administrative Distance

RIPv2

OSPFv2

EIGRP

EIGRP for IPv6

OSPFv3

IS-IS

IS-IS for IPv6

BGP

BGP-MP

EGP

load balancing

equal-cost load balancing

unequal-cost load balancing